Decision Brief: Scaling Remote Farmer for Multi-Session, Multi-Sprint, and Repo Cartography

Date: 2026-03-12 | **Audience**: Solo developer, team leads | **Phase**: Compiled

Executive Summary

Remote Farmer should scale via three independent workstreams: (1) session-keyed server state with `Map<session_id, SessionState>` for multi-session support (r003, p003), (2) a `sprints/` directory model with `wheat.config.json` pointer for multi-sprint (r020, r025, p004), and (3) a compiler-generated topic-map manifest for repo cartography (r011, r017, p001). All three prototypes are built and tested. The manifest generator produces results in <10ms (p001). Total implementation effort: 5-9 hours across all workstreams (r008, r026).

Recommendation

Multi-Session (13 claims, tested evidence)

Use a single-server, session-keyed architecture (r003). Replace global singletons with a `Map<session_id, SessionState>` and a `getSession(id)` lazy initializer (r004). Claude Code's hook payloads already include a stable `session_id` UUID (r001), and SessionStart/SessionEnd lifecycle hooks provide explicit session boundaries (r002, w001).

Critical constraint: SessionStart/SessionEnd only support command-type hooks, not HTTP (w002) -- a curl wrapper is required for Farmer integration.

Prototype validated: SessionState class, session-keyed Maps, SSE tagging, and pill-bar session switcher with hue-from-hash color coding are implemented and tested (p003). Backwards compatible with single-session via 'default' fallback.

Dashboard UX: unified feed with session color-coding, pill-bar session switcher (r005, p002). Permission cards must prominently display session labels to mitigate cross-session approval confusion (r006). Heartbeat timeout (5min) for crashed sessions (r007). Estimated effort: 2-4 hours (r008).

Multi-Sprint (9 claims, tested evidence)

Use a `sprints/` directory with one subdirectory per sprint, each self-contained (r020). The current sprint pointer lives in `wheat.config.json` (not symlinks) for cross-platform portability (r025, resolved against r020). Dashboard shows a collapsible sprint card list for switching (r022). Cross-sprint claim references use `sprint-slug:claim-id` format (r024). Server adds `--sprints-dir` flag with hot-reload via `fs.watchFile` (r021).

Prototype validated: Sprint scanning, `/api/sprints` endpoint, hot-reload, and backwards-compatible `--claims` flag are implemented and tested (p004).

Sprint lifecycle: `/init` creates under `sprints/`, archiving sets `meta.phase` to 'archived' (r023). Estimated effort: 3-5 hours (r026).

Repo Cartography (8 claims, tested evidence)

Generate `wheat-manifest.json` as a topic map on every compilation (r011, r017). The manifest maps topics to claim IDs, file paths, and sprints -- enabling a single `Read` call to replace 3-5 iterative Glob/Grep searches (r012). CLAUDE.md remains the behavioral instruction layer; the manifest handles structural discovery (r015). Commit messages should include `[topic]` tags for fast `git log --grep` (r014).

Prototype validated: Generator built and tested at <10ms, ~5KB output (p001).

Cartography is needed now: the repo has 157 files (x001), not 80 as previously estimated (r016 challenged). The `examples/` directory creates 62 duplicate-named files causing search noise (x002). Context window pollution -- not I/O time -- is the real cost metric (x003).

Compatibility

Zero npm dependencies constraint maintained (d004). All implementations use Node built-in modules only. Single-session workflow preserved via backwards-compatible defaults.

Evidence Summary

Topic	Claims	Max Evidence	Sources	Types	Key Finding
multi-session	13	tested	4	5/6	SessionState Maps + session_id routing, prototype working (p003)

multi-sprint	9	tested	3	5/6	sprints/ dir + config pointer, prototype working (p004)
cartography	8	tested	4	4/6	Manifest generator built, <10ms, 5KB (p001)
performance	6	tested	4	4/6	157 files now, context pollution is real cost (x001-x003)
compatibility	1	stated	1	1/6	Zero deps constraint, not independently verified

Tradeoffs and Risks

Risk	Evidence	Mitigation
Concurrent permission confusion across sessions (r006)	stated	Color-coded session labels, distinct hue per session_id
CLAUDE.md staleness if used for structural cartography (r015)	documented	Use manifest as structural truth, CLAUDE.md for behavior only
Symlink portability on Windows (r025)	documented	Config file as canonical pointer, symlinks as convenience only
SessionStart/SessionEnd command-only hooks (w002)	documented	Curl wrapper scripts instead of direct HTTP hooks
Context window pollution from noisy search results (x003)	documented	Manifest eliminates iterative search, saves 10-40K tokens/session

Blind Spots (from /blind-spot analysis)

- **Security**: No claims address attack vectors for Farmer exposed via Cloudflare Tunnel
- **Dashboard UX**: UX decisions scattered across topics, no dedicated risk analysis
- **Compatibility**: Single constraint claim (d004), never independently tested
- **Corroboration**: Zero claims have independent corroboration from a second source

Resolved Conflicts

r020 vs r025 (multi-sprint): r020 recommended symlinks for current-sprint pointer. r025 identified Windows portability risk. Resolution: use `wheat.config.json`

`currentSprint` field as canonical pointer; symlinks optional as local convenience, not committed to git. Both claims remain active with complementary guidance.

Implementation Roadmap

Phase	Workstream	Effort	Status	Files
1	Manifest generator	--	DONE (p001)	generate-manifest.js
2	Multi-session server refactor	2-4h (r008)	DONE (p003)	server.mjs, dashboard.html
3	Multi-sprint support	3-5h (r026)	DONE (p004)	server.mjs, dashboard.html, wheat-init.js, wheat-compiler.js
4	Compiler manifest integration	1h	TODO	wheat-compiler.js

Appendix: Claim Inventory

ID	Type	Topic	Evidence	Content
d001	constraint	multi-session	stated	Multi-session dashboard support required
d002	constraint	multi-sprint	stated	Multi-sprint viewing and switching required
d003	constraint	cartography	stated	Repo must be self-describing for AI sessions
d004	constraint	compatibility	stated	Zero npm deps, backwards compatible
d005	constraint	performance	stated	Measurably faster search with cartography
r001	factual	multi-session	documented	session_id UUID in all hook payloads
r002	factual	multi-session	documented	SessionStart/SessionEnd lifecycle hooks

r003	recommendation	multi-session	documented	Single-server, session-keyed Maps
r004	recommendation	multi-session	documented	getSession(id) lazy initializer
r005	recommendation	multi-session	documented	Unified feed with session color-coding
r006	risk	multi-session	stated	Concurrent permission confusion
r007	factual	multi-session	documented	Heartbeat timeout for stale sessions
r008	estimate	multi-session	stated	~300 lines, 2-4 hours
r010	factual	cartography	documented	CLAUDE.md auto-loads per directory
r011	recommendation	cartography	documented	wheat-manifest.json topic map
r012	factual	performance	documented	Single Read replaces 3-5 tool calls
r013	recommendation	cartography	documented	Sprint index in compilation output
r014	recommendation	cartography	documented	[topic] tags in commit messages
r015	risk	cartography	documented	Per-directory CLAUDE.md staleness
r016	estimate	performance	stated	Crossover at 150-200 files (CHALLENGED by x001)
r017	recommendation	cartography	documented	Topic map over file tree
x001	factual	performance	tested	Repo already 157 files, not 80
x002	factual	performance	tested	examples/ creates 62 duplicate-name files
x003	risk	performance	documented	Context pollution is real cost, not I/O
w001	factual	multi-session	documented	SessionStart/SessionEnd confirmed with source field

w002	factual	multi-session	documented	Lifecycle hooks are command-only, not HTTP
r020	recommendation	multi-sprint	documented	sprints/ directory + config pointer
r021	recommendation	multi-sprint	documented	--sprints-dir flag with hot-reload
r022	recommendation	multi-sprint	documented	Collapsible card list sprint switcher
r023	recommendation	multi-sprint	documented	/init creates under sprints/, phase-based lifecycle
r024	recommendation	multi-sprint	documented	Cross-sprint refs: sprint-slug:claim-id
r025	risk	multi-sprint	documented	Symlink portability, use config instead
r026	estimate	multi-sprint	stated	~250 lines, 3-5 hours
p001	factual	cartography	tested	Manifest generator: <10ms, 5KB topic map
p002	recommendation	multi-session	documented	Multi-session refactor plan with SessionState
p003	factual	multi-session	tested	Multi-session prototype implemented and working
p004	factual	multi-sprint	tested	Multi-sprint prototype implemented and working

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